

Detailed Technical Report (V2): Robust Multi-View Human Reconstruction with Specialized Hand Fusion

Technical Documentation

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Abstract

This document describes the implemented V2 pipeline for multi-view human reconstruction. The system combines a three-stage architecture with an optional external specialized hand model workflow: (i) per-view SAM-3D inference and hand-keypoint fusion, (ii) robust triangulation with inlier-gated refinement, and (iii) rig-constrained optimization with quality gates, bad-frame handling, and sequence-level recovery/smoothing. The report focuses on the current code behavior, interfaces, objective functions, and runtime controls.

1 Pipeline Summary

The current pipeline is orchestrated by a sequence runner and consists of:

1. **Stage 0 (optional):** specialized hand precompute in a separate environment.
2. **Stage 1:** per-view SAM-3D inference, optional hand fusion in 2D.
3. **Stage 2:** robust multi-view triangulation on an MHR subset.
4. **Stage 3:** MHR optimization to triangulated 3D targets.

Sequence-level logic includes:

- quality classification (`ok`, `bad_loss`, failure statuses),
- optional retry for bad optimization (default retry count is zero),
- recovery for missing/bad frames by interpolation or bounded copy,
- optional smoothing and sequence summary export.

2 Data Layout and Stage Contracts

2.1 Input Layout

For frame index t and camera c , expected image layout is:

$$\text{image_folder}/t/c.\text{jpg} \tag{1}$$

with at least two cameras per valid frame.

2.2 Main Outputs

Under `output_root`:

- `inference/npz/...`: per-view Stage-1 dictionaries.
- `triangulation/.../triangulated.npz`: Stage-2 robust 3D subset.
- `optimization/.../opt_out.npy`: Stage-3 optimized output.
- `optimization/.../opt_out_smoothed.npy`: optional smoothed output.
- `sequence_summary.json`: per-frame status and metrics.

2.3 Inter-Stage Contract

- Stage-1 to Stage-2: `pred_keypoints_2d` with shape $(70, 2)$ or $(N, 70, 2)$.
- Stage-2 to Stage-3: `points3d_refined` $(M, 3)$, `subset_indices`, `inlier_mask`, per-camera re-projection diagnostics.
- Stage-3 output: optimized SAM-style dictionary and `opt_*` diagnostics.

3 Stage 0: External Specialized Hand Precompute

When specialized hand models are not available inside the SAM environment, the system supports external precompute.

- Script: `run_wilor_precompute.py`.
- Per-image output: `<out>/<rel_dir>/<image_stem>.npy`.
- Each file stores `detections` with fields: `is_right`, `hand_bbox`, `pred_keypoints_2d` $(21, 2)$.

This decouples environments:

- WiLoR environment runs hand inference.
- SAM pipeline environment consumes saved detections.

4 Stage 1: SAM Inference with Hand Fusion

4.1 Per-View Base Inference

For each image $I_t^{(c)}$, SAM-3D predicts:

$$\mathbf{z}_t^{(c)} = f_{\theta}(I_t^{(c)}), \quad (2)$$

including 2D keypoints, 3D keypoints, vertices, and rig-related parameters.

4.2 Hand Fusion Inputs

Let $\mathbf{u}_{t,c,k}^{\text{sam}} \in \mathbb{R}^2$ be SAM 2D keypoint $k \in \{0, \dots, 69\}$. Let $\mathbf{h}_{t,c,j}^{\text{sp}} \in \mathbb{R}^2$ be specialized hand keypoint $j \in \{0, \dots, 20\}$.

Mapping functions convert specialized 21-joint order to MHR70 right/left indices:

$$m_R(j), m_L(j) \in \{0, \dots, 69\}. \quad (3)$$

4.3 Candidate Selection per Side

For each side (right/left), among candidate detections:

$$d = \|\mathbf{h}_{\text{wrist}}^{\text{sp}} - \mathbf{u}_{\text{wrist}}^{\text{sam}}\|_2. \quad (4)$$

The selected candidate minimizes d under wrist-distance gating:

$$d < \tau_{\text{wrist}}. \quad (5)$$

If SAM wrist is invalid, a fallback heuristic prefers larger hand box area.

4.4 Replacement Rule

If a side candidate is accepted, mapped hand points replace SAM hand points:

$$\mathbf{u}_{t,c,m(j)} = \begin{cases} \mathbf{h}_{t,c,j}^{\text{sp}}, & \text{if finite and accepted,} \\ \mathbf{u}_{t,c,m(j)}^{\text{sam}}, & \text{otherwise.} \end{cases} \quad (6)$$

Wrist replacement is optional (`replace_wrist_with_specialized`).

The stage also writes source metadata:

- `pred_keypoints_2d_source` (length 70 mask),
- `specialized_hand_status_right/left`,
- `specialized_hand_num_points`, `specialized_hand_used`.

4.5 Fusion Debug Visualization

When enabled, Stage-1 saves overlays that compare:

- SAM hand points before fusion,
- replaced specialized points,
- fallback points retained from SAM.

This is intended for quick visual validation of per-frame fusion quality.

5 Stage 2: Robust Multi-View Triangulation

5.1 Supervised Subset

Triangulation uses a high-utility MHR subset:

- right-hand keypoints (20),
- left-hand keypoints (20),
- shoulders, elbows, wrists (6),

so $M = 46$ points per frame.

5.2 Camera Model

For camera v :

$$\mathbf{x}_{\text{cam}}^{(v)} = R_v \mathbf{X} + t_v, \quad \mathbf{u}^{(v)} = \pi_v(\mathbf{X}; K_v, D_v). \quad (7)$$

DLT initialization uses undistorted normalized rays; residuals are evaluated in pixels through full projection.

5.3 Step A: Pairwise Candidate Generation

For each valid camera pair, generate $\mathbf{X}$ by linear triangulation (DLT).

5.4 Step B: Robust Candidate Scoring

For candidate $\mathbf{X}$, per-view reprojection residual magnitude is:

$$r_v(\mathbf{X}) = \left\| \mathbf{u}^{(v)} - \pi_v(\mathbf{X}) \right\|_2. \quad (8)$$

Aggregate score options:

$$S_{\text{median}}(\mathbf{X}) = \text{median}\{r_v\}, \quad (9)$$

$$S_{\text{trimmed}}(\mathbf{X}) = \frac{1}{K-1} \sum_{k=1}^{K-1} r_{(k)}, \quad (10)$$

$$S_{\text{huber}}(\mathbf{X}) = \sum_v \rho_\delta(r_v), \quad (11)$$

with Huber

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & r \leq \delta, \\ \delta \left(r - \frac{1}{2}\delta \right), & r > \delta. \end{cases} \quad (12)$$

Select candidate with minimum robust score. If pair candidates fail, fall back to all-view DLT.

5.5 Step C: Inlier View Gating

From selected initialization $\mathbf{X}^{\text{init}}$:

$$\mathcal{I}_{j,t} = \left\{ v \mid \left\| \mathbf{u}_{j,t}^{(v)} - \pi_v(\mathbf{X}_{j,t}^{\text{init}}) \right\|_2 < \tau \right\}. \quad (13)$$

If fewer than two inliers are found but multiple views exist, force the two smallest-error views. Optional reseed recomputes DLT from inliers.

5.6 Step D: Inlier-Only LM Refinement

For each point:

$$\mathbf{X}_{j,t}^* = \arg \min_{\mathbf{X}} \sum_{v \in \mathcal{I}_{j,t}} \left\| \mathbf{u}_{j,t}^{(v)} - \pi_v(\mathbf{X}) \right\|_2^2. \quad (14)$$

Levenberg–Marquardt update:

$$\Delta \mathbf{X} = -(J^\top J + \lambda I)^{-1} J^\top \mathbf{r}, \quad \mathbf{X} \leftarrow \mathbf{X} + \Delta \mathbf{X}. \quad (15)$$

Optional robust LM applies residual weighting with Huber-style weights.

5.7 Triangulation Output

`triangulated.npz` contains:

- `points3d_refined`, `subset_indices`, `subset_names`,
- `inlier_mask` (M, V),
- per-camera reprojection errors (init and refined),
- robust-scoring and inlier-threshold metadata.

6 Stage 3: Rig-Constrained Optimization

6.1 Variables and Forward Model

State decomposition:

$$\boldsymbol{\theta}_t = (\mathbf{p}_t, \mathbf{h}_t, \mathbf{s}_t, \mathbf{b}_t, \mathbf{e}_t), \quad (16)$$

where:

- $\mathbf{p}_t$: body pose parameters,
- $\mathbf{h}_t$: hand108 parameters,
- $\mathbf{s}_t, \mathbf{b}_t, \mathbf{e}_t$: scale, shape, expression blocks.

Forward kinematics produces landmarks:

$$\hat{\mathbf{Y}}_{t,m} = \mathcal{F}_m(\boldsymbol{\theta}_t), \quad m \in \mathcal{M}. \quad (17)$$

6.2 View-Based Initialization

Each view provides an initialization candidate. Candidates are ranked by weighted aligned 3D residual to triangulated targets, with triangulation mean pixel error as tie-break.

6.3 Fixed Non-Pose Parameters

For single-person sequences, non-pose blocks can be fixed from a reference frame/camera:

$$\mathbf{s}_t \equiv \mathbf{s}_{\text{ref}}, \quad \mathbf{b}_t \equiv \mathbf{b}_{\text{ref}}, \quad \mathbf{e}_t \equiv \mathbf{e}_{\text{ref}}. \quad (18)$$

Hand block can also be fixed if explicitly enabled.

6.4 Masking and Lower-Body Freeze

Binary mask $\mathbf{m}$ defines trainable dimensions:

$$\mathbf{p}_t^{\text{eff}} = \mathbf{p}_t \odot \mathbf{m} + \mathbf{p}_t^{\text{fix}} \odot (1 - \mathbf{m}). \quad (19)$$

Hand/jaw-related dimensions are masked by default; optional lower-body freeze extends the frozen set. Frozen dimensions are re-clamped every step.

6.5 Aligned Data Loss

Similarity transform aligns prediction to triangulated targets:

$$\bar{\mathbf{Y}}_{t,m} = s_t R_t \hat{\mathbf{Y}}_{t,m} + t_t. \quad (20)$$

By default, alignment is estimated from stable anchors (shoulders, elbows, wrists, optional neck/acromion), while loss is computed on all valid supervised points.

Data term:

$$\mathcal{L}_{\text{data},t} = \frac{\sum_{m \in \mathcal{M}} w_{t,m} \rho_{\delta} \left(\left\| \bar{\mathbf{Y}}_{t,m} - \mathbf{Y}_{t,m} \right\|_2 \right)}{\sum_{m \in \mathcal{M}} w_{t,m} + \epsilon}. \quad (21)$$

Weights $w_{t,m}$ are derived from triangulation inlier reliability (`inlier_mask` averaged across views), then sanitized with finite-point checks.

6.6 Total Objective

$$\mathcal{L}_t = \mathcal{L}_{\text{data},t} \lambda_{\text{pose}} \mathcal{L}_{\text{pose},t} \lambda_{\text{temp}} \mathcal{L}_{\text{temp},t} \lambda_{\text{vel}} \mathcal{L}_{\text{vel},t} \lambda_{\text{acc}} \mathcal{L}_{\text{acc},t} \lambda_{\text{hand}} \mathcal{L}_{\text{hand},t}, \quad (22)$$

with:

$$\mathcal{L}_{\text{pose},t} = \left\| \mathbf{p}_t - \mathbf{p}_t^0 \right\|_2^2, \quad (23)$$

$$\mathcal{L}_{\text{temp},t} = \left\| \mathbf{p}_t - \mathbf{p}_{t-1} \right\|_2^2, \quad (24)$$

$$\mathcal{L}_{\text{vel},t} = \left\| \mathbf{p}_t - \hat{\mathbf{p}}_t^{\text{vel}} \right\|_2^2, \quad (25)$$

$$\mathcal{L}_{\text{acc},t} = \left\| (\mathbf{p}_t - \mathbf{p}_{t-1}) - (\mathbf{p}_{t-1} - \mathbf{p}_{t-2}) \right\|_2^2, \quad (26)$$

$$\mathcal{L}_{\text{hand},t} = \left\| \mathbf{h}_t - \mathbf{h}_t^0 \right\|_2^2. \quad (27)$$

In current defaults, $\lambda_{\text{vel}} = \lambda_{\text{acc}} = 0$, so velocity and acceleration terms are inactive unless explicitly enabled.

6.7 Optimization Procedure

- Optimizer: Adam on trainable pose (and optional hand108).
- Stabilization: gradient clipping and frozen-gradient zeroing.
- Stops: minimum-iteration gate, patience on non-improvement, and divergence ratio.
- Best iterate (lowest total loss) is restored before final export.

6.8 Bad-Frame Classification

Each frame stores:

- `best_loss`, `final_loss`,
- `best_data_loss`, `final_data_loss`.

A frame is marked bad if any condition holds:

$$\text{best_loss} > \tau_{\text{loss}}, \quad \text{best_data_loss} > \tau_{\text{data}}, \quad \frac{\text{final_loss}}{\max(\text{best_loss}, \epsilon)} > \gamma, \quad \frac{\text{final_data_loss}}{\max(\text{best_data_loss}, \epsilon)} > \gamma. \quad (28)$$

7 Sequence Orchestration and Recovery

7.1 Frame Discovery and Stage Control

The orchestrator discovers frame folders, enforces a minimum-view requirement, and can skip/reuse stage artifacts.

7.2 Temporal Guard

Temporal priors are disabled after a long non-good streak:

$$\text{stale_run} \geq \text{max_stale_temporal_frames}. \quad (29)$$

This avoids stale-state lock-in.

7.3 Bad-Frame Retry

Retry logic exists but is optional. Current default is:

$$\text{bad_frame_max_retries} = 0, \quad (30)$$

so retries are disabled unless explicitly set.

7.4 Recovery Policy

Bad/missing optimization outputs are treated as unusable and replaced by:

1. interpolation when valid neighbors exist on both sides,
2. bounded one-sided copy when only one side exists and distance is within `max_edge_recovery_copy_span`.

Recovered frames are explicitly labeled as recovered and their copied optimization metrics are cleared to avoid misleading reports.

7.5 Optional Sequence Smoothing

Post-processing can apply nearest-valid fill, temporal median filtering, outlier suppression, and bidirectional EMA on key numeric fields.

8 Important Runtime Controls

Control	Effect
<code>enable_specialized_hand_fusion</code>	Enables hand keypoint replacement in Stage-1.
<code>specialized_hand_source</code>	<code>precomputed</code> or <code>live</code> .
<code>specialized_hand_input_root</code>	Root of external hand detections for precomputed mode.
<code>specialized_hand_wrist_max_dist_px</code>	Wrist-distance gate for accepting specialized candidates.
<code>specialized_hand_debug_vis</code>	Saves Stage-1 SAM-vs-specialized fusion overlays.
<code>score_type, huber_delta</code>	Robust scoring mode for triangulation candidate selection.
<code>inlier_thresh</code>	Reprojection threshold for inlier view gating.
<code>robust_lm, robust_lm_delta</code>	Enables robust residual weighting in LM refinement.
<code>fixed_mhr_param_frame_idx,</code> <code>fixed_mhr_param_cam</code>	Fixes non-pose blocks from one reference frame/camera.
<code>fixed_hand_pose_from_reference</code>	Also fixes hand pose from reference if enabled.
<code>optimize_hand_pose</code>	Enables hand108 optimization in Stage-3.
<code>use_anchor_similarity</code>	Estimates similarity from stable anchors instead of all supervised points.
<code>freeze_lower_body</code>	Freezes lower-body dimensions to reduce stationary jitter.
<code>bad_frame_max_retries</code>	Optional bad-frame retries (default 0).
<code>max_edge_recovery_copy_span</code>	Bounds one-sided recovery copy distance.

9 Recommended V2 Configuration (Static Single-Person)

1. Use precomputed specialized hand detections and fuse in Stage-1.
2. Fix non-pose identity blocks from a clean reference frame (front camera).
3. Keep anchor-only similarity enabled.
4. Enable lower-body freeze for stationary standing sequences.
5. Keep velocity/acceleration weights at zero unless a specific motion scenario justifies them.
6. Keep bad-frame retries disabled and rely on interpolation recovery.

10 Reproducibility Commands

10.1 External Hand Precompute

```
python run_wilor_precompute.py \  
  --image_folder /path/to/frames_root \  
  --output_root /path/to/wilor_precomputed \  
  --device cuda \  
  --hand_conf 0.3 \  
  --rescale_factor 2.5 \  

```


`--debug_vis`

10.2 End-to-End Pipeline

```
python run_full_pipeline.py \  
  --image_folder /path/to/frames_root \  
  --output_root /path/to/pipeline_out \  
  --cams left front right \  
  --caliscope_toml /path/to/config.toml \  
  --mhr_py mhr70.py \  
  --checkpoint_path /path/to/model.ckpt \  
  --mhr_path /path/to/mhr_model.pt \  
  --hf_repo facebook/sam-3d-body-dinov3 \  
  --enable_specialized_hand_fusion \  
  --specialized_hand_source precomputed \  
  --specialized_hand_model wilor \  
  --specialized_hand_input_root /path/to/wilor_precomputed \  
  --specialized_hand_debug_vis \  
  --fixed_mhr_param_frame_idx 10 \  
  --fixed_mhr_param_cam front \  
  --freeze_lower_body
```

11 Limitations

- Accuracy depends on camera calibration and synchronization quality.
- Hand fusion quality depends on specialized detector robustness under occlusion and blur.
- Subset-based supervision may under-constrain body parts not represented in the triangulated subset.
- Recovery is a fallback for robustness, not a substitute for high-quality observations.

12 Conclusion

The implemented V2 pipeline provides a practical and controllable reconstruction workflow: specialized hand cues improve 2D evidence quality, robust triangulation improves 3D supervision reliability, and rig-constrained optimization enforces anatomical consistency with explicit diagnostics and recovery. The main strength is sequence robustness with transparent frame-level status semantics.

A Overleaf Usage

This file is LaTeX source stored as `paper_detail_v2.md`. For Overleaf:

1. upload `paper_detail_v2.md`,
2. rename it to `main.tex`,
3. compile with pdfLaTeX.